

CI/CD Using Azure Pipelines

CICD | 1 Day



Build, test, and ship continuously with Azure Pipelines. This one-day course is designed to help agile teams configure and use Azure Pipelines to practice Continuous Integration, Continuous Delivery, and Continuous Deployment, from your first build pipeline to releasing with feature flags and learning from production. You'll work in teams in a shared Azure DevOps project, building and deploying a common case study the way a real team would.

Who should take this course?

This course is for anyone on a software development team, especially those involved in automating the building, testing, and deploying of software. It's equally valuable for people outside the development team, including managers, IT, operations, and other stakeholders, who want hands-on exposure to what Azure Pipelines can do. Some experience with Azure DevOps is recommended, and familiarity with C# and .NET is beneficial but not required.

Course content

This course progressively builds your fluency with Azure Pipelines, from your first build pipeline to releasing software with feature flags and learning from production. You'll work hands-on in teams in a shared Azure DevOps project, building, deploying, and experimenting on a common case study the way a real team would.

1. CONTINUOUS INTEGRATION

- Introduction to Azure Pipelines
- Creating a build pipeline
- Pipeline tasks, variables, and triggers
- Microsoft-hosted vs. self-hosted agents
- Building pipelines as code with YAML
- Running automated tests in the pipeline
- Collecting and reporting code coverage
- Practicing Continuous Integration
- Hands-on lab

2. CONTINUOUS DELIVERY

- Creating a release pipeline
- Pipeline stages, jobs, tasks, and targets
- Multi-stage YAML pipelines
- Configuring environments and resources
- Creating, deploying, and managing a release
- Practicing Continuous Delivery
- Hands-on lab

3. CONTINUOUS DEPLOYMENT

- Continuous Delivery vs. Continuous Deployment
- Build, measure, learn
- Feature flags as a kill switch, for targeting, and for experiments
- Hypothesis-Driven Development (HDD)
- Telemetry and learning from production
- A/B testing and experimenting in production
- Managing feature flags
- Hands-on lab

Course Designer

This course was designed by Richard Hundhausen, Microsoft's first Visual Studio ALM/DevOps MVP, a Professional Scrum Trainer, and co-creator of the Nexus framework, with decades helping teams build and ship software. To see other Azure DevOps and GitHub courses, visit www.accentient.com.